Array

An array is a type of linear data structure that is defined as a collection of elements with same or different data types. They exist in both single dimension and multiple dimensions. These data structures come into picture when there is a necessity to store multiple elements of similar nature together at one place.

Memory Address	2391	2392	2393	2394	2395
Array Values	12	34	68	77	43
Array Index	0	1	2	3	4

The difference between an array index and a memory address is that the array index acts like a key value to label the elements in the array. However, a memory address is the starting address of free memory available.

Following are the important terms to understand the concept of Array.

- **Element** − Each item stored in an array is called an element.
- **Index** − Each location of an element in an array has a numerical index, which is used to identify the element.

Syntax

Creating an array in **C** and **C++** programming languages

```
data_type array_name[array_size]={elements separated by commas}
or,
data_type array_name[array_size];
```

Need for Arrays

Arrays are used as solutions to many problems from the small sorting problems to more complex problems like travelling salesperson problem. There are many data structures other than arrays that provide efficient time and space complexity for these problems, so what makes using arrays better? The answer lies in the random access lookup time.

Arrays provide **O(1)** random access lookup time. That means, accessing the 1st index of the array and the 1000th index of the array will both take the same time. This is due to the fact that array comes with a pointer and an offset value. The pointer points to the right location of the memory and the offset value shows how far to look in the said memory.

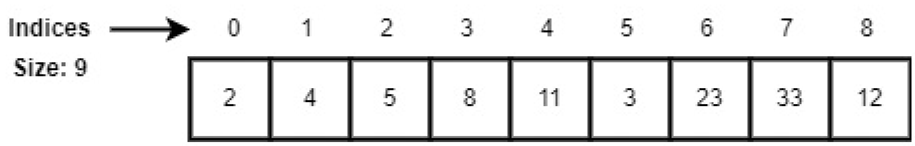
```
array_name[index]
   |        |
Pointer   Offset
```

Therefore, in an array with 6 elements, to access the 1st element, array is pointed towards the 0th index. Similarly, to access the 6th element, array is pointed towards the 5th index.

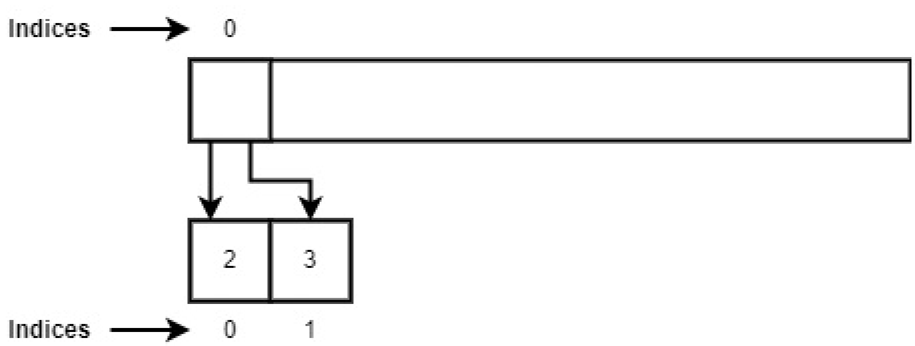
Array Representation

Arrays are represented as a collection of buckets where each bucket stores one element. These buckets are indexed from '0' to 'n-1', where n is the size of that particular array. For example, an array with size 10 will have buckets indexed from 0 to 9.

This indexing will be similar for the multidimensional arrays as well. If it is a 2-dimensional array, it will have sub-buckets in each bucket. Then it will be indexed as array_name[m][n], where m and n are the sizes of each level in the array.



Single Dimensional Array



Multi Dimensional Array

As per the above illustration, following are the important points to be considered.

- Index starts with 0.
- Array length is 9 which means it can store 9 elements.
- Each element can be accessed via its index. For example, we can fetch an element at index 6 as 23.

Basic Operations in Arrays

The basic operations in the Arrays are insertion, deletion, searching, display, traverse, and update. These operations are usually performed to either modify the data in the array or to report the status of the array.

Following are the basic operations supported by an array.

- **Traverse** − print all the array elements one by one.
- **Insertion** − Adds an element at the given index.
- **Deletion** − Deletes an element at the given index.
- **Search** − Searches an element using the given index or by the value.
- **Update** − Updates an element at the given index.
- **Display** − Displays the contents of the array.

Array - Insertion Operation

In the insertion operation, we are adding one or more elements to the array. Based on the requirement, a new element can be added at the beginning, end, or any given index of array. This is done using input statements of the programming languages.

Algorithm

Following is an algorithm to insert elements into a Linear Array until we reach the end of the array

```
1. Start
2. Create an Array of a desired datatype and size.
3. Initialize a variable 'i' as 0.
4. Enter the element at ith index of the array.
5. Increment i by 1.
6. Repeat Steps 4 & 5 until the end of the array.
7. Stop
```

Example

```
#include <iostream>        // For input and output
using namespace std;       // So we don’t need to write std:: again and again

int main() {               // Program starts here

   int LA[3] = {}, i;      // Array of size 3, all values set to 0, and variable i

   cout << "Array Before Insertion:" << endl;    // Message before array values

   for(i = 0; i < 3; i++)    // Loop to show array values
      cout << "LA[" << i << "] = " << LA[i] << endl; // Print each element

   cout << "Inserting elements.." << endl;      // Message for insertion
   cout << "Array After Insertion:" << endl;    // Message after insertion

   for(i = 0; i < 5; i++) { // Loop to insert values
      LA[i] = i + 2;         // Store value (i + 2) in array
      cout << "LA[" << i << "] = " << LA[i] << endl; // Print new value
   }

   return 0;                 // End of program
}
```

Output

```
Array Before Insertion:
LA[0] = 0
LA[1] = 0
LA[2] = 0
Inserting elements..
Array After Insertion:
LA[0] = 2
LA[1] = 3
LA[2] = 4
LA[3] = 5
LA[4] = 6
```

Array - Deletion Operation

In this array operation, we delete an element from the particular index of an array. This deletion operation takes place as we assign the value in the consequent index to the current index.

Algorithm

Consider LA is a linear array with N elements and K is a positive integer such that K<=N. Following is the algorithm to delete an element available at the K^{th} position of LA.

```
1. Start
2. Set J = K
3. Repeat steps 4 and 5 while J < N
4. Set LA[J] = LA[J + 1]
5. Set J = J+1
6. Set N = N-1
7. Stop
```

- **Start**

→ Begin the algorithm

- **Set J = K**

→ Store the position of the element to be deleted in J

- **Repeat steps 4 and 5 while J < N**

→ Loop runs to shift elements to the left

- **Set LA[J] = LA[J + 1]**

→ Move the next element into the current position

- **Set J = J + 1**

→ Move to the next index

- **Set N = N − 1**

→ Reduce the size of the array by 1

- **Stop**

→ End the algorithm

```
#include <iostream>         // For input and output
using namespace std;        // To use cout without std::

int main() {                // Program starts here

   int LA[] = {1, 3, 5};    // Declare and initialize array
   int i, n = 3;            // i for loop, n stores size of array

   cout << "The original array elements are :" << endl; // Print message

   for(i = 0; i < n; i++) {         // Loop to print original array
      cout << "LA[" << i << "] = " << LA[i] << endl;
   }

   for(i = 1; i < n; i++) {         // Loop to delete element
      LA[i] = LA[i + 1];            // Shift elements to the left
      n = n - 1;                    // Reduce array size
   }
```

```
   cout << "The array elements after deletion :" << endl; // Print message

   for(i = 0; i < n; i++) {         // Loop to print updated array
      cout << "LA[" << i << "] = " << LA[i] << endl;
   }
}
```

Output

```
The original array elements are :
LA[0] = 1
LA[1] = 3
LA[2] = 5
The array elements after deletion :
LA[0] = 1
LA[1] = 5
```

Array - Search Operation

Searching an element in the array using a key; The key element sequentially compares every value in the array to check if the key is present in the array or not.

Algorithm

Consider LA is a linear array with N elements and K is a positive integer such that K<=N. Following is the algorithm to find an element with a value of ITEM using sequential search.

```
1. Start
2. Set J = 0
3. Repeat steps 4 and 5 while J < N
4. IF LA[J] is equal ITEM THEN GOTO STEP 6
5. Set J = J +1
6. PRINT J, ITEM
7. Stop
```

- **Start**

→ Begin the algorithm

- **Set J = 0**

→ Start searching from the first position of the array

- **Repeat steps 4 and 5 while J < N**

→ Loop through all array elements

- **IF LA[J] is equal to ITEM THEN GOTO STEP 6**

→ Check if the current element matches the item to be searched

- **Set J = J + 1**

→ Move to the next element

- **PRINT J, ITEM**

→ Display the position and value of the found item

- **Stop**

→ End the algorithm

```cpp
#include <iostream>         // For input and output
using namespace std;        // To use cout without std::

int main() {                // Program starts here

   int LA[] = {1, 3, 5, 7, 8}; // Declare and initialize array
   int item = 5, n = 5;       // item to search, n is size of array
   int i = 0;                 // Loop variable

   cout << "The original array elements are : " << endl; // Print message

   for(i = 0; i < n; i++) {  // Loop to print array elements
      cout << "LA[" << i << "] = " << LA[i] << endl;     // Print each element
   }
```

```cpp
   for(i = 0; i < n; i++) {  // Loop to search the item
      if(LA[i] == item) {    // Check if item is found
         cout << "Found element " << item
              << " at position " << i + 1 << endl; // Print position
      }
   }

   return 0;                 // End of program
}
```

Output

```
The original array elements are :
LA[0] = 1
LA[1] = 3
LA[2] = 5
LA[3] = 7
LA[4] = 8
Found element 5 at position 3
```

Array - Traversal Operation

This operation traverses through all the elements of an array. We use loop statements to carry this out.

Algorithm

Following is the algorithm to traverse through all the elements present in a Linear Array

```
1  Start
2. Initialize an Array of certain size and datatype.
3. Initialize another variable i with 0.
4. Print the ith value in the array and increment i.
5. Repeat Step 4 until the end of the array is reached.
6. End
```

- **Start**

→ Begin the algorithm

- **Initialize an array of certain size and data type**

→ Create an array to store values

- **Initialize variable `i` with 0**

→ Start from the first index of the array

- **Print the ith value of the array and increment `i`**

→ Display the current element and move to the next index

- **Repeat Step 4 until the end of the array is reached**

→ Continue until all elements are printed

- **End**

→ Stop the algorithm

```cpp
#include <iostream>        // For input and output
using namespace std;       // To use cout without std::

int main() {               // Program starts here

   int LA[] = {1, 3, 5, 7, 8};  // Declare and initialize array
   int item = 10, k = 3, n = 5; // item to use later, position k, and size of array
   int i = 0, j = n;            // Loop variables

   cout << "The original array elements are:\n"; // Message

   for(i = 0; i < n; i++)       // Loop to print array elements
      cout << "LA[" << i << "] = " << LA[i] << endl; // Print each element

   return 0;                     // End of program
}
```

Output

```
The original array elements are :
LA[0] = 1
LA[1] = 3
LA[2] = 5
LA[3] = 7
LA[4] = 8
```

Array - Update Operation

Update operation refers to updating an existing element from the array at a given index.

Algorithm

Consider LA is a linear array with N elements and K is a positive integer such that K<=N. Following is the algorithm to update an element available at the Kth position of LA.

```
1. Start
2. Set LA[K-1] = ITEM
3. Stop
```

- **Start**

→ Begin the algorithm

- **Set LA[K-1] = ITEM**

→ Store the item at position `K` in the array (`K-1` because array index starts from 0)

- **Stop**

→ End the algorithm

```
int main() {                // Program starts here

   int LA[] = {1, 3, 5, 7, 8};  // Declare and initialize array
   int item = 10, k = 3, n = 5; // item to insert, position k, size of array
   int i = 0, j = n;            // Loop variables

   cout << "The original array elements are :\n"; // Message before update

   for(i = 0; i < n; i++)       // Loop to print original array
      cout << "LA[" << i << "] = " << LA[i] << endl; // Print each element

   LA[2] = item;                // Update element at position 3 (index 2) with item

   cout << "The array elements after updation are :\n"; // Message after update

   for(i = 0; i < n; i++)       // Loop to print updated array
      cout << "LA[" << i << "] = " << LA[i] << endl;   // Print each element

   return 0;                     // End of program
}
```

Output

```
The original array elements are :
LA[0] = 1
LA[1] = 3
LA[2] = 5
LA[3] = 7
LA[4] = 8
The array elements after updation :
LA[0] = 1
LA[1] = 3
LA[2] = 10
LA[3] = 7
LA[4] = 8
```

Array - Display Operation

This operation displays all the elements in the entire array using a print statement.

Algorithm

Consider LA is a linear array with N elements. Following is the algorithm to display an array elements.

```
1. Start
2. Print all the elements in the Array
3. Stop
```

```cpp
#include <iostream>        // For input and output
using namespace std;      // To use cout without std::

int main() {              // Program starts here

   int LA[] = {1, 3, 5, 7, 8};  // Declare and initialize array
   int n = 5;                  // Size of the array
   int i;                       // Loop variable

   cout << "The original array elements are :\n"; // Message before printing

   for(i = 0; i < n; i++)       // Loop to go through all array elements
      cout << "LA[" << i << "] = " << LA[i] << endl; // Print each element

   return 0;                    // End of program
}
```

Output

```
The original array elements are :
LA[0] = 1
LA[1] = 3
LA[2] = 5
LA[3] = 7
LA[4] = 8
```